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Subject: EDS

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CODETANTRA

1.1.1. Calculate Momentum 04:41

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

Test case 1

5.0

10.0

50.00kgm/s

Test case 2

2.3

4.8

11.04kgm/s

Explorer

calculateM...

Submit

Debugger

```python
mass = float(input())
velocity = float(input())
momentum = mass * velocity
print(f"{momentum:.2f}kgm/s")
```

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Test case 1

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4.8

11.04kgm/s

Explorer

calculateM...

Submit

Debugger

```
mass = float(input())
velocity = float(input())
momentum = mass * velocity
print(f"{momentum:.2f}kgm/s")
```

Average time
0.031 s
30.75 ms

Maximum time
0.039 s
39.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 39 ms Debug

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

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CODETANTRA

1.1.2. Conditional Calculation Based on the ... 03:52

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:

$1 \le n \le 999$

Input Format:

- The input consists of a single integer n.

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

Test case 1

9

81

Test case 2

166

5.50

Test case 3

24

4.90

Test case 4

3456

Invalid

Explorer

conditional...

Submit

Debugger

```python
import math

number = int(input())

if 1 <= number <= 9:
    print(number ** 2)

elif 10 <= number <= 99:
    print(f"{math.sqrt(number):.2f}")

elif 100 <= number <= 999:
    print(f"{number ** (1/3):.2f}")

else:
    print("Invalid")
```

Average time 0.017 s 17.29 ms

Maximum time 0.021 s 21.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 21 ms Debug

Expected output: 9 / 81

Actual output: 9 / 81

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CODETANTRA

1.1.3. Days between Two Dates 15:40

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

Test case 1

2014-07-02
2014-11-02
123

Test case 2

2014-07-02
2014-07-11
9

noOfDays.py

Submit

```python
from datetime import date
from datetime import datetime

date1=input().strip()
date2=input().strip()

d1=datetime.strptime(date1,"%Y-%m-%d")
d2=datetime.strptime(date2,"%Y-%m-%d")

difference=d2-d1

print(difference.days)
```

Average time
0.038 s
38.25 ms

Maximum time
0.064 s
64.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 64 ms Debug

Expected output	Actual output
2014-07-02	2014-07-02
2014-11-02	2014-11-02
123	123

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1.1.4. Reverse a Number 02:06

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

Test case 1

5367

7635

Test case 2

0132

231

Explorer

reverseNu...

Submit

```
#write your code here...
num = int(input())
reversed_num = int(str(num)[::-1])
print( reversed_num)
```

Debugger

Average time 0.022 s 21.80 ms

Maximum time 0.027 s 27.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 20 ms Debug

Expected output	Actual output
5367	5367
7635	7635

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1.1.5. Multiplication Table

00:39

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

Test case 1

```
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
```

Test case 2

```
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Explorer

multiplicat...

Submit

Debugger

```python
#write your code here...
num = int(input())

for i in range(1,11):
    print(f"{num} x {i} = {num * i}")
```

Average time	Maximum time
0.018 s	0.025 s
18.25 ms	25.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 18 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16

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1.2.1. Pass or Fail

00:17

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

Test case 1

```
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division
```

Test case 2

```
5
56 78 97 86 93
Aggregate Percentage: 82.00
Grade: Distinction
```

Test case 3

```
5
99 85 43 97 34
Fail
```

Test case 4

```
3
55 54 53
Aggregate Percentage: 54.00
```

Explorer

passorFail....

Submit

```python
# write your code here
def cal(marks, total_courses):
    if any(mark < 40 for mark in marks):
        return "Fail"

    total_marks = sum(marks)
    aggregate_percentage = (total_marks / (total_courses * 100)) * 100

    if aggregate_percentage > 75:
        grade = "Distinction"
    elif aggregate_percentage >= 60:
        grade = "First Division"
    elif aggregate_percentage >= 50:
        grade = "Second Division"
    elif aggregate_percentage >= 40:
        grade = "Third Division"

    return (aggregate_percentage, grade)


num_courses = int(input())
marks = list(map(int, input().split()))

result = cal(marks, num_courses)

if result == "Fail":
    print("Fail")
else:
    aggregate_percentage, grade = result
    print(f"Aggregate Percentage: {aggregate_percentage:.2f}")
    print(f"Grade: {grade}")
```

Debugger

Average time	Maximum time
0.029 s	0.042 s
28.90 ms	42.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 42 ms Debug

Expected output	Actual output
4	4
55 65 78 89	55 65 78 89
Aggregate Percentage: 71.75	Aggregate Percentage: 71.75

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1.2.2. Fibonacci Series 14:33

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

Test case 1

5

0 1 1 2 3

Test case 2

15

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377

Explorer

fibonacci.py

Submit

```python
def fibonacci(n):

    if n==1:
        return 0
    elif n==2:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)
n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")

```

Debugger

Average time
0.036 s
35.75 ms

Maximum time
0.082 s
82.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 20 ms Debug

Expected output	Actual output
5	5
0 1 1 2 3	0 1 1 2 3

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1.2.3. Pattern - 1 21:40

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

Test case 1

```
5
*
* *
* * *
* * * *
* * * * *
```

Test case 2

```
4
*
* *
* * *
* * * *
```

Explorer

rightangle... Submit

```python
#Type Content here...
n=int(input())
for i in range(1,n+1):
    for j in range(i):
        print("*",end=" ")
    print()
```

Debugger

Average time 0.018 s 17.83 ms

Maximum time 0.026 s 26.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 26 ms Debug

Expected output	Actual output
5	5
*	*
* *	* *

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1.2.4. Pattern - 2 03:09

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

Test case 1

```
5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Test case 2

```
8
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
```

Explorer

numberPat... Submit

```
# Type Content here...
n=int(input())
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

Debugger

Average time 0.026 s 25.75 ms

Maximum time 0.043 s 43.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 43 ms Debug

Expected output	Actual output
5	5
1	1
1 2	1 2

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